

Name..... Adm. No..... Sec.....

1. If the three quartiles of a distribution are 142, 160 and 185, comment on the shape of the distribution.

A)symmetric B)negatively skew ☒ C) positively skew D)can not be said.

mark: 1

2. The first three moments of a distribution about the value 3 of the variable are 2, 10, 30 respectively. The first three moments about zero are respectively

☒ A) 5, 31, 201 B)5, 25, 175 C)10, 65, 100 D)5, 35, 205

mark: 2

3. The standard deviation of a symmetrical distribution is 5. The value of the fourth central moment in order that the distribution becomes mesokurtic is:

☒ A) 1875 B) 1850 C) 1900 D) 1925

mark: 1

4. Let the joint PDF of random variables X and Y be

$$f(x, y) = \begin{cases} 6e^{-(2x+3y)}, & x \geq 0, y \geq 0 \\ 0 & \text{otherwise.} \end{cases}.$$

Then $E[Y|X \geq 2]$ and $P(X \geq Y)$ are

A) $\frac{1}{2}$ and 2 B) $\frac{1}{2}$ and $\frac{2}{5}$ ☒ C) $\frac{1}{3}$ and $\frac{3}{5}$ D) $\frac{2}{3}$ and $\frac{3}{5}$

mark: 2

5. Suppose a continuous r.v. X has the MGF $M_X(t) = (1 - 2t)^{-\frac{1}{2}}$ for $t \leq \frac{1}{2}$. The mean and variance of X is

A) 2 and 2 B) 3 and 5 C) 1 and 3 ☒ D) 1 and 2

mark: 2

6. An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T, the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0 & \text{if } t < 1; \\ \frac{1}{4} & \text{if } 1 \leq x < 3; \\ \frac{1}{2} & \text{if } 3 \leq x < 5; \\ \frac{3}{4} & \text{if } 5 \leq x < 7; \\ 1 & \text{if } x \geq 7. \end{cases}$$

Find $P(T \leq 5|T \geq 2)$

☒ A) $\frac{2}{3}$ B) $\frac{1}{3}$ C) $\frac{1}{2}$ D) $\frac{3}{4}$

mark: 2